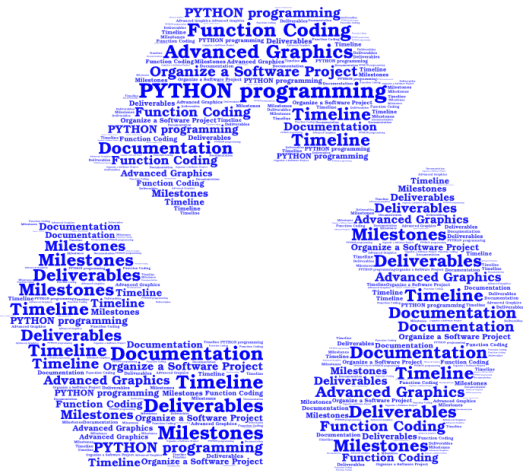




# Project Work: Calculating Results Preparing the Final Report

Lecture: Modelling an Simulation

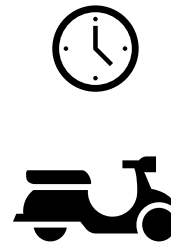
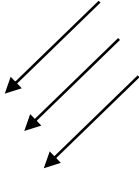
This is a compulsory elective module  
of  
**Technology of Circular  
Economy**  
(Master of Engineering)

**Dr. Uwe Reimer**  
Senior Scientist

# Project Goal

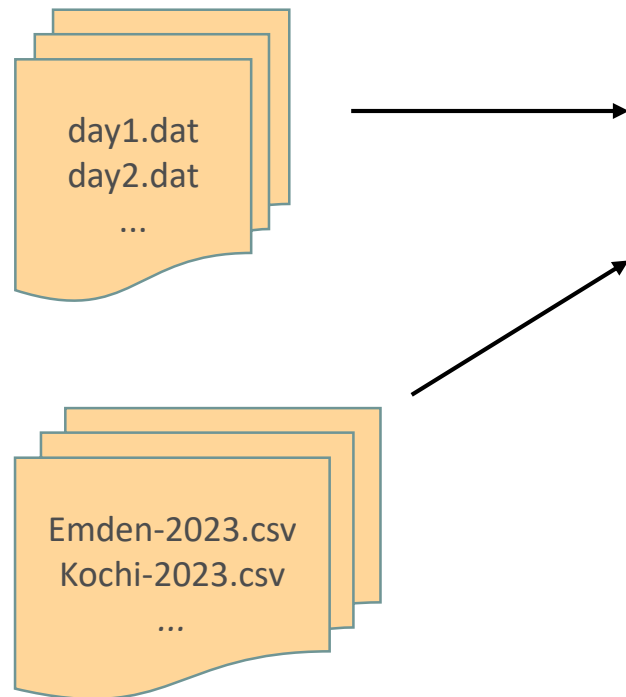
Design a PV system which collects energy from the sun and stores it in a battery. An electric scooter shall be charged with this energy for commuting to work (daily drive cycle).

Results shall be presented for the city of Emden and second location in a different country.



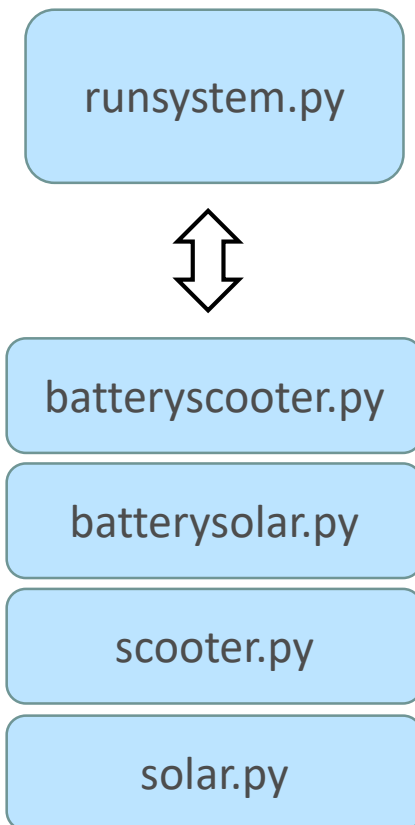
## Input Files:

- Files for the driving cycle
- PV Power for the city



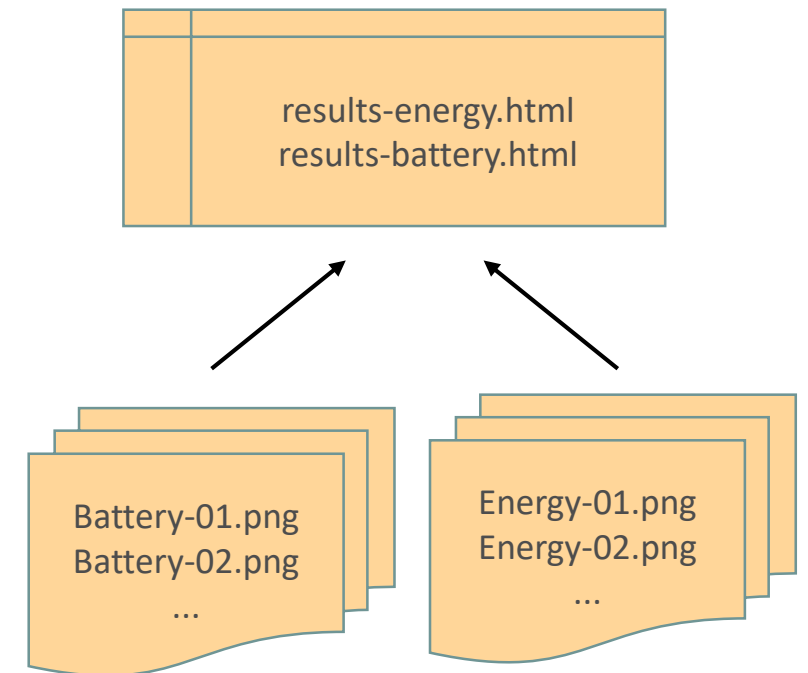
## Input Data:

- Choice of city
- Size of PV module



## Output Files:

- HTML file for energy and battery
- \*.PNG files for energy and battery



## Which input data is available?

### Scooter driving cycle:

7 days with distances of 13.6 km and 14.5 km.

| Day | Power | Distance |
|-----|-------|----------|
| 0   | 109.3 | 13.7     |
| 1   | 109.3 | 13.7     |
| 2   | 116.0 | 14.5     |
| 3   | 109.3 | 13.7     |
| 4   | 109.3 | 13.7     |
| 2   | 116.0 | 14.5     |
| 6   | 109.3 | 13.7     |

### PV production:

9 cities with PV system (slope 23°, orientation south).

The data is obtained from [https://re.jrc.ec.europa.eu/pvg\\_tools/en/](https://re.jrc.ec.europa.eu/pvg_tools/en/) for the year 2023.

### Cities are:

- Emden, London, Paris, Oslo (Europe)
- Kochi, Bangalore, Hyderabad, Mumbai (India)
- Accra (Ghana)
- Beijing (China)

## How to run the program:

- 1) You can either use a batch file with the line `'python runsystem.py > output.txt'` or
- 2) Open the Windows PowerShell, navigate to the correct directory and type `'python runsystem.py'`.

## Steps to be taken:

- 1) Open the file `'runsystem.py'` in NotePad++
- 2) In line 13 set the size of PV system.
- 3) In line 20 set the number for the correct city.
- 4) Run the program.
- 5) Check the HTML pages `'results-energy.html'` and `'results-battery.html'` for output.  
(Remember to reload the pages, if new output was generated.)

## Question 1):

What is the minimum size of a PV system in order to drive the scooter for a whole year?  
Choose the city of Emden and at least one other city.

## Question 2):

For the results of Question 1 -> What is the total cost of driving the scooter – calculated per year?  
Assume the following prices:

- Scooter = 350 EUR (incl. battery)
- Battery = 350 EUR (for PV, size 480 Wh identical to scooter)
- PV panel = 1 EUR / 1 Watt peak power

The cost (CAPEX) are distributed over 10 years, i.e. divide the sum by 10 years to obtain the driving cost.

## 1 Software: Task and Solution

### 1.1 Introduction

### 1.2 Software Requirements

### 1.3 Software design and workflow

### 1.4 Final results: Emden

### 1.5 Final results: Hometown

*your own documentation  
(see lecture slides for help)*

*your results*

## 2 Project Management

### 2.1 Project Plan

### 2.2 Risk analysis

### 2.3 Monitoring and control

### 2.4 Deliverables and Milestones

*Check the lecture slides for the date  
when Deliverables and Milestones  
have been reached.*

## 3 Feedback for the lecture

If you want, you can write down your experience, suggestions and any feedback.

My intentions with this lecture have been:

- |                           |                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b><i>Programming</i></b> | You can do it. You know how to use a programming language to open a data file, do some calculations and create a plot. |
| <b>Project Plan</b>       | You know how to plan work for yourself and others. The key is to break down big problems into small steps.             |

# This is the end

I hope, you found this lecture useful in some ways.

If yes, then go on and start to create ...

😊 Uwe Reimer, July 2026

